Emergent Constitutions: A Heterogeneous-Agent Model with Endogenous Governance and LLM-Augmented Political Decision Making

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Abstract

We develop a dynamic stochastic general equilibrium model with heterogeneous agents (DSGE-HA) in which governance institutions are fully endogenous. Agents choose between working and entrepreneurship via utility-based occupational choice, face idiosyncratic productivity and ability shocks discretized via the Rouwenhorst method, and collectively design taxation, redistribution, public goods provision, regulation, and voting rules through a constitutional framework. Economic decisions (consumption, savings, labor supply) are solved via value function iteration on the Bellman equation, while political decisions—proposal generation and voting—are delegated to large language models (LLMs) that receive structured economic context, mimicking bounded rationality in institutional design. Analytical Cobb-Douglas market clearing yields zero-error equilibrium prices each period. The mutable constitution feeds back into economic equilibrium through fiscal policy and market regulation. We characterize the recursive competitive equilibrium, describe the computational algorithm, and discuss the model's implications for the co-evolution of institutions and inequality.

Keywords: Heterogeneous agents, endogenous institutions, occupational choice, LLM agents, constitutional dynamics, value function iteration

JEL Classification: D72, E21, H11, L26, C63

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1 Introduction

A central question in political economy is how institutions emerge, persist, and evolve. Standard macroeconomic models take institutions—tax codes, property rights, voting rules—as exogenous parameters. Yet real-world governance is the product of collective action by heterogeneous agents whose economic circumstances shape their political preferences and vice versa. This paper develops a computational framework that endogenizes the entire governance layer within a heterogeneous-agent macroeconomic model.

1.1 Motivation

The canonical incomplete-markets models of Aiyagari [3], Huggett [6], and Krusell and Smith [8] have yielded deep insights into wealth inequality, precautionary savings, and aggregate dynamics under idiosyncratic risk. However, these models typically hold fiscal policy fixed: the tax rate, transfer schedule, and public goods provision are parameters, not outcomes. In reality, agents vote on these policies, propose new rules, and form coalitions to advance their interests. The feedback loop—from economic inequality to political action to institutional change to new equilibrium outcomes—is precisely what this paper formalizes.

1.2 Contributions

This paper makes three contributions.

First, we build a DSGE-HA model with occupational choice between workers and entrepreneurs, endogenous firm dynamics with R&D-driven productivity growth, and Cobb-Douglas production at the firm level. Idiosyncratic productivity and entrepreneurial ability follow AR(1) processes discretized via the Rouwenhorst method [7], while aggregate TFP follows a separate AR(1) process. Household decisions are solved via value function iteration (VFI) on the Bellman equation with a discrete leisure grid.

Second, we introduce a mutable constitutional framework with eight rule types—tax schedules, transfer programs, public goods provision, market regulation, firm regulation, voting procedures, property rights, and custom rules—each with sandboxed enforcement code. Constitutional rules feed back into agent budget constraints through taxation, transfers, and regulation. Agents propose, validate, vote on, and amend rules through a formal political process in which voting thresholds themselves are endogenous.

Third, we integrate large language models (LLMs) into the political decision layer. Economic optimization is analytically well-defined (the Bellman equation has a known structure), so we solve it numerically. Political deliberation, by contrast, is open-ended: what tax rate to propose, how to frame a regulation, whether a means-tested transfer is better than a universal one. We delegate these decisions to LLMs that receive structured context about the economy and society, producing proposals and votes that reflect bounded rationality in institutional design. A benchmark mode with purely numerical governance enables controlled comparisons.

1.3 Related Literature

Our model builds on several strands of the literature. The heterogeneous-agent tradition originates with Bewley [4], Huggett [6], and Aiyagari [3], with Krusell and Smith [8] extending to aggregate shocks. Cagetti and De Nardi [5] introduce entrepreneurship into the Bewley framework. We follow their approach of modeling occupational choice with heterogeneous entrepreneurial ability but add endogenous governance and LLM-driven institutional dynamics.

On the computational side, our discretization of AR(1) processes follows Kopecky and Suen [7], who show that the Rouwenhorst method dominates Tauchen's method for highly persistent processes. Our VFI implementation uses exponentially-spaced asset grids and linear interpolation, standard techniques in the quantitative macro toolkit.

The endogenous institutions literature includes Acemoglu and Robinson [2], who study the dynamics of political institutions and economic outcomes, and Acemoglu [1], who formalize the relationship between political power and economic institutions. Our model operationalizes these ideas computationally: agents with heterogeneous preferences over equality and liberty vote on the very rules that govern economic outcomes.

The use of LLMs as economic agents is nascent. We contribute by showing how LLM-generated political decisions can be integrated into a fully specified general equilibrium model while preserving determinism (temperature zero, seeded RNG) and providing analytical fallback (benchmark mode).

The remainder of the paper is organized as follows. Section 2 describes the model environment. Section 3 formalizes the constitutional framework. Section 4 defines the recursive competitive equilibrium. Section 5 details the computational method. Section 6 discusses the LLM integration. Section 7 presents the calibration. Section 8 discusses results and implications. Section 9 concludes.

2 Model Environment

2.1 Time, Agents, and State Space

Time is discrete, indexed by $t = 0, 1, 2, \ldots, T$. The economy is populated by $N \geq 20$ agents, indexed by $i = 1, \ldots, N$.

Definition 1 (Individual State). *Each agent i at time t is characterized by the individual state vector*

$$s_{it} = (a_{it},\, z_{it},\, e_{it},\, o_{it}), \tag{1}$$

where $a_{it} \in [\underline{a}, \infty)$ denotes assets (wealth), $z_{it} \in \mathcal{S}_z$ is idiosyncratic labor productivity, $e_{it} \in \mathcal{S}_e$ is entrepreneurial ability, and $o_{it} \in \{W, E\}$ denotes occupation (worker or entrepreneur).

Definition 2 (Aggregate State). *The aggregate state at time t is*

$$\Gamma_t = (\mu_t,\, \mathcal{C}_t,\, w_t,\, r_t,\, A_t), \tag{2}$$

where μ_t is the cross-sectional distribution of individual states, $\mathcal{C}_t$ is the constitution (a mutable set of rules), w_t is the wage, r_t is the interest rate, and A_t is aggregate total factor productivity (TFP).

2.2 Preferences

Agents have Cobb-Douglas preferences over private consumption c, leisure ℓ, and per-capita public goods G:

$$u(c, \ell, G) = c^{\alpha_u} \cdot \ell^{\beta_u} \cdot G^{\gamma_u}, \qquad \alpha_u + \beta_u + \gamma_u = 1, \tag{3}$$

where $\alpha_u, \beta_u, \gamma_u > 0$ are the weights on consumption, leisure, and public goods, respectively. The constraint $\alpha_u + \beta_u + \gamma_u = 1$ ensures homogeneity of degree one.

The lifetime objective is

$$\max\ \mathbb{E}_0 \sum_{t=0}^{\infty} \beta_d^t\, u(c_{it}, \ell_{it}, G_t), \tag{4}$$

where $\beta_d \in (0, 1)$ is the subjective discount factor.

Remark 1 (Preference Heterogeneity). *The model supports two modes. Under* homogeneous preferences*, all agents share the same $(\alpha_u, \beta_u, \gamma_u, \beta_d)$, enabling a significant computational speedup: VFI is solved once and policies are interpolated per agent. Under* heterogeneous preferences, *each agent draws utility weights from a Dirichlet distribution and* β_d *from* $U[0.9, 0.99]$, *requiring per-agent VFI.*

In addition to economic preferences, each agent possesses an ideological value vector $v_i = (v_i^{eq}, v_i^{lib})$ with $v_i^{eq} + v_i^{lib} = 1$, where v_i^{eq} represents the preference weight toward equality and v_i^{lib} toward liberty. These values influence political decisions—agents with high v_i^{eq} tend to favor redistribution, while those with high v_i^{lib} favor low taxes and free markets.

2.3 Technology

2.3.1 Workers

A worker with productivity z_{it} and leisure choice $\ell_{it} \in [0, 1]$ supplies effective labor

$$h_{it} = z_{it} \cdot (1 - \ell_{it}), \tag{5}$$

and earns labor income $w_t \cdot h_{it}$.

2.3.2 Entrepreneurs and Firms

An entrepreneur i owns a firm f that produces output according to the Cobb-Douglas production function

$$Y_f = A_f \cdot K_f^{\alpha} \cdot L_f^{1-\alpha}, \tag{6}$$

where $A_f = e_{it} \cdot A_t$ is firm-specific TFP (the product of the entrepreneur's ability and aggregate TFP), K_f is the firm's capital, L_f is effective labor hired, and $\alpha \in (0, 1)$ is the capital share.

The first-order condition for optimal labor demand is

$$(1 - \alpha) \cdot A_f \cdot K_f^{\alpha} \cdot L_f^{-\alpha} = w_t, \tag{7}$$

yielding optimal labor

$$L_f^* = \left(\frac{(1 - \alpha) \cdot A_f \cdot K_f^{\alpha}}{w_t} \right)^{1/\alpha}. \tag{8}$$

Firm profit is

$$\pi_f = Y_f - w_t L_f^* - (r_t + \delta) K_f, \tag{9}$$

where δ is the depreciation rate and $r_t + \delta$ is the rental rate of capital.

2.3.3 R&D and Firm-Specific TFP Growth

Active firms may invest in R&D at expenditure x_f^{rd}. The probability of a TFP improvement is

$$p_f = p_0 \cdot \sqrt{\frac{x_f^{rd}}{\bar{x}^{rd}}}, \tag{10}$$

where p_0 is the base success probability and $\bar{x}^{rd}$ is the mean R&D spending across all active firms. Upon success, firm TFP updates as

$$A'_f = A_f \cdot (1 + \eta), \qquad \eta \sim \max\left\{0, \mathcal{N}(\mu_{rd}, \sigma_{rd}^2)\right\}. \tag{11}$$

2.4 Occupational Choice: Entry and Exit

Occupational choice is determined by comparing lifetime utility as a worker versus an entrepreneur. This utility-based framework replaces profit-based entry/exit conditions, capturing the full cost of entrepreneurship including reduced leisure.

2.4.1 Firm Value

The value of a firm to an entrepreneur with ability e and capital K uses a closed-form infinite-horizon formula based on the stationary distribution of the ability Markov chain:

$$V(e, K) = \pi(e, K) + \frac{\beta_f \cdot \pi(\bar{e}, K)}{1 - \beta_f}, \tag{12}$$

where $\pi(e, K)$ is current-period profit at actual ability e, $\pi(\bar{e}, K)$ is the stationary expected profit evaluated at $\bar{e} = \mathbb{E}_{\pi^*}[e]$ (the mean ability under the stationary distribution π^* of the ability Markov chain), and β_f is the firm discount factor. The first term reflects exact current-period profit; the second term treats future profits as a perpetuity at the stationary expected level.

The stationary distribution π^* is computed once at initialization via power iteration on the ability transition matrix Π_e, converging to $\|\pi^{(k+1)} - \pi^{(k)}\|_\infty < 10^{-12}$.

2.4.2 Worker Lifetime Value

The lifetime utility value of remaining a worker is approximated as a perpetuity of per-period utility at the household's optimal allocation:

$$V^W(s_i) = \frac{u(c_W^*, \ell_W^*, G)}{1 - \beta_d}, \tag{13}$$

where $\ell_W^* \approx 0.35$ is the typical optimal leisure from VFI, and $c_W^* \approx 0.7 \times y_{avail}$ is approximate optimal consumption (consuming roughly 70% of available income including labor and asset income).

2.4.3 Entrepreneur Lifetime Value

The lifetime utility value of entrepreneurship accounts for firm profits, reduced leisure from running a firm, and the entry cost penalty:

$$V^E(s_i) = \frac{u(c_E^*, \ell_E, G)}{1 - \beta_d} - \Phi(F), \tag{14}$$

where $\ell_E = 0.2$ reflects the reduced leisure of an entrepreneur (running a firm takes effort), c_E^* is consumption from firm profits plus asset income, and $\Phi(F)$ is the entry cost expressed in utility units:

$$\Phi(F) = F \cdot \frac{\alpha_u}{c_E^*} \cdot u(c_E^*, \ell_E, G), \tag{15}$$

i.e., the entry cost F scaled by the marginal utility of consumption. The term $\Phi(F)$ is included only for new entrants, not for continuing entrepreneurs.

2.4.4 Entry Condition

A worker enters entrepreneurship if two conditions hold:

$$V^E(s_i) > V^W(s_i) \qquad \text{and} \qquad a_{it} \geq \underline{K} + F, \tag{16}$$

where $\underline{K}$ is the minimum capital requirement and F is the sunk entry cost. The first condition ensures that entrepreneurship yields higher lifetime utility; the second ensures financial feasibility.

2.4.5 Exit Condition

An existing entrepreneur exits (reverts to worker) if

$$V^W(s_i) > V^E(s_i), \tag{17}$$

where V^E is evaluated without the entry cost penalty (since the entrepreneur is already operating). Exit occurs when lifetime worker utility exceeds the ongoing value of the firm.

2.4.6 Optimal Capital

The optimal capital level K^* maximizes firm value subject to feasibility:

$$K^* = \arg \max_{K \in [\underline{K}, a_{it} - F]} V(e_i, K), \tag{18}$$

solved via grid search over 20 candidate capital levels.

2.5 Stochastic Processes

The model features three sources of uncertainty.

2.5.1 Idiosyncratic Productivity

Labor productivity follows an AR(1) process in logs:

$$\log z_{it} = \rho_z \log z_{i,t-1} + \varepsilon_{it}^z, \qquad \varepsilon_{it}^z \sim \mathcal{N}(0, \sigma_z^2). \tag{19}$$

2.5.2 Entrepreneurial Ability

Entrepreneurial ability follows an independent AR(1) process:

$$\log e_{it} = \rho_e \log e_{i,t-1} + \varepsilon_{it}^e, \qquad \varepsilon_{it}^e \sim \mathcal{N}(0, \sigma_e^2). \tag{20}$$

2.5.3 Aggregate TFP

Aggregate TFP follows an AR(1) process:

$$\log A_t = \rho_A \log A_{t-1} + \varepsilon_t^A, \qquad \varepsilon_t^A \sim \mathcal{N}(0, \sigma_A^2). \tag{21}$$

2.5.4 Rouwenhorst Discretization

The continuous AR(1) processes (19) and (20) are discretized into finite-state Markov chains using the Rouwenhorst method [7]. For a process with persistence ρ and innovation volatility σ:

Step 1. Compute the unconditional standard deviation: $\sigma_z^{unc} = \sigma / \sqrt{1 - \rho^2}$.

Step 2. Set grid bounds: $z_{max} = \sigma_z^{unc} \cdot \sqrt{n - 1}$, where n is the number of grid points.

Step 3. Construct an equally-spaced log grid: $\{-z_{max}, -z_{max} + \Delta, \ldots, z_{max}\}$ with $\Delta = 2z_{max}/(n - 1)$.

Step 4. Set the transition probability parameter: $p = (1+\rho)/2$.

Step 5. Build the transition matrix recursively. The base case ($n = 2$) is

$$\Pi_2 = \begin{pmatrix} p & 1-p \\ 1-p & p \end{pmatrix}. \tag{22}$$

For $n > 2$, construct Π_n from Π_{n-1} via four-quadrant expansion:

$$\widetilde{\Pi}_n = p \begin{pmatrix} \Pi_{n-1} & \mathbf{0} \\ \mathbf{0}^\top & 0 \end{pmatrix} + (1-p) \begin{pmatrix} \mathbf{0} & \Pi_{n-1} \\ 0 & \mathbf{0}^\top \end{pmatrix} + (1-p) \begin{pmatrix} \mathbf{0}^\top & 0 \\ \Pi_{n-1} & \mathbf{0} \end{pmatrix} + p \begin{pmatrix} 0 & \mathbf{0}^\top \\ \mathbf{0} & \Pi_{n-1} \end{pmatrix}, \tag{23}$$

followed by row normalization: $\Pi_n(i,\cdot) = \widetilde{\Pi}_n(i,\cdot)/\sum_j \widetilde{\Pi}_n(i,j)$.

Step 6. Convert the log grid to levels: $\mathcal{S} = \{\exp(z_1), \ldots, \exp(z_n)\}$.

The stationary distribution π^* of Π is computed by power iteration until $\|\pi^{(k+1)} - \pi^{(k)}\|_\infty < 10^{-12}$.

2.6 Markets and Budget Constraints

2.6.1 Factor Market Clearing

In equilibrium, factor markets clear. Labor market clearing requires

$$\sum_{f \in \mathcal{F}_t} L_f^*(w_t, r_t) = \sum_{i: o_{it} = W} z_{it} \cdot (1 - \ell_{it}), \tag{24}$$

where the left-hand side is aggregate labor demand from firms and the right-hand side is aggregate effective labor supply from workers.

Capital market clearing requires

$$\sum_{f \in \mathcal{F}_t} K_f = \sum_{i=1}^{N} a_{it}, \tag{25}$$

where the left-hand side is aggregate capital demand (firm capital) and the right-hand side is aggregate savings.

2.6.2 Worker Budget Constraint

A worker faces the budget constraint

$$a_{i,t+1} = (1 + r_t)a_{it} + w_t z_{it}(1 - \ell_{it}) - c_{it} - T(y_{it}) + Tr_{it}, \tag{26}$$

where $T(y_{it})$ is the tax schedule evaluated at labor income $y_{it} = w_t z_{it}(1 - \ell_{it})$, and Tr_{it} is transfers received. Agents face the borrowing constraint $a_{i,t+1} \geq \underline{a}$.

2.6.3 Entrepreneur Budget Constraint

An entrepreneur additionally receives firm profit:

$$a_{i,t+1} = (1 + r_t)a_{it} + w_t z_{it}(1 - \ell_{it}) + \pi_{f(i)} - c_{it} - T(y_{it}) + Tr_{it}, \tag{27}$$

where $\pi_{f(i)}$ is the profit of the firm owned by agent i.

2.6.4 Goods Market Identity

By Walras' law, the goods market clears:

$$Y_t = C_t + I_t + G_t, \tag{28}$$

where $Y_t = \sum_f Y_f$ is aggregate output, $C_t = \sum_i c_{it}$ is aggregate consumption, $I_t = \sum_f \delta K_f$ is replacement investment, and G_t is government (public goods) spending.

3 Governance and Constitutional Framework

3.1 Constitutional State Space

The constitution at time t is a finite collection of rules:

$$\mathcal{C}_t = \{R_1, R_2, \ldots, R_M\}, \tag{29}$$

where each rule $R_m = (name_m, type_m, \theta_m, f_m(\cdot))$ consists of a unique identifier, a type from the set

$$\mathcal{T} = \{\text{tax\_schedule, transfer\_program, public\_goods, market\_regulation, firm\_regulation, voting\_proce} \tag{30}$$

a parameter vector θ_m (e.g., $\theta = \{rate : 0.2\}$ for a tax rule), and an enforcement function $f_m(\cdot)$ specified as a sandboxed expression.

3.2 Fiscal Policy

3.2.1 Taxation

The tax on agent i's income is determined by all active rules of type `tax_schedule`:

$$T_i = \sum_{m:type_m=\text{tax\_schedule}} f_m(y_i; \theta_m), \tag{31}$$

subject to the non-negativity constraint $T_i \in [0, y_i]$ (taxes cannot exceed income). The default rule is a flat tax: $T_i = \tau \cdot y_i$ with $\tau \in [0, 1]$.

Total tax revenue is $R = \sum_{i=1}^{N} T_i$.

3.2.2 Public Goods

Public goods spending is determined by rules of type `public_goods`:

$$G_t = \frac{\phi \cdot R}{N}, \tag{32}$$

where $\phi \in [0, 1]$ is the fraction of revenue allocated to public goods, specified by the constitutional rule parameters.

3.2.3 Transfers

The remaining revenue $R - \phi R$ is distributed according to transfer rules. Three transfer methods are available.

Under *equal-share* redistribution:

$$Tr_i = \frac{(1-\phi)R}{N}. \tag{33}$$

Under *means-tested* redistribution:

$$Tr_i = \frac{(1-\phi)R \cdot \omega_i}{\sum_{j=1}^{N} \omega_j}, \qquad \omega_i = \frac{1}{\max\{a_{it}, 1\}}, \tag{34}$$

so that poorer agents receive larger transfers.

Under *progressive* redistribution, the same inverse-wealth weighting as means-tested is used:

$$Tr_i^{prog} = \frac{(1-\phi)R \cdot \omega_i}{\sum_{j=1}^{N} \omega_j}, \qquad \omega_i = \frac{1}{\max\{a_{it}, 1\}}. \tag{35}$$

The progressive method serves as an alternative label for the inverse-wealth weighting scheme, available as a distinct policy choice in the constitutional framework.

3.3 Political Process

The constitutional amendment process follows four stages.

Stage 1. **Proposal.** At proposal-interval periods (every K_{prop} periods), each agent may propose one constitutional change. A proposal specifies an action $\in$ {add, modify, remove}, a rule name, a rule type (for additions), parameter values, and optionally enforcement code.

Stage 2. **Validation.** Each proposal is validated against structural constraints:

- Tax rates must lie in $[0, 1]$.
- Transfer methods must be recognized (equal-share, means-tested, progressive).
- Public goods fractions must lie in $[0, 1]$.
- Enforcement code must pass an AST-level safety check (restricted to arithmetic, comparisons, and safe builtins; no imports, function definitions, or attribute access).
- Additions cannot duplicate existing rule names; modifications require existing rules; the active voting rule cannot be removed.

Invalid proposals are rejected before reaching the voting stage.

Stage 3. **Voting.** Each agent casts a vote (for or against) on each validated proposal. The outcome is determined by the currently active voting procedure rule with threshold $\bar{\theta} \in (0, 1]$:

$$\text{Proposal passes} \iff \begin{cases} V_{for} > N \cdot \bar{\theta}, & \text{if } \bar{\theta} \leq 0.5 \text{ (majority)}, \\ V_{for} \geq \lceil N \cdot \bar{\theta} \rceil, & \text{if } 0.5 < \bar{\theta} < 1 \text{ (supermajority)}, \\ V_{for} = N, & \text{if } \bar{\theta} = 1 \text{ (unanimity)}, \end{cases} \tag{36}$$

where V_{for} is the number of votes in favor and N is the total number of eligible voters. Ties are resolved in favor of the status quo.

Stage 4. **Amendment.** Passed proposals are applied to the constitution in order. Each application re-validates the resulting rule before activation. The constitution is immutable within a period and only updated between periods.

3.4 Endogenous Feedback

The distinguishing feature of this framework is the feedback loop between governance and economics. Constitutional rules determine:

- The tax function $T(\cdot)$ that enters budget constraints (26)–(27);
- The transfer schedule Tr_i that enters household income;
- Public goods G_t that enters the utility function (3);
- Market and firm regulations that constrain economic behavior.

These in turn affect prices (w_t, r_t), wealth dynamics $a_{i,t+1}$, occupational choice, and ultimately the political preferences that generate new proposals. The constitution is thus endogenous in the deepest sense: it is both a determinant and a consequence of economic equilibrium.

4 Equilibrium Definition

4.1 Worker's Problem

A worker with state (a, z), facing prices (w, r), public goods G, tax function $T(\cdot)$, and transfer Tr, solves the Bellman equation

$$V^W(a, z) = \max_{c,\ell} \left\{ u(c, \ell, G) + \beta_d \, \mathbb{E}\big[V(a', z') \,\big|\, z \big] \right\} \tag{37}$$

subject to

$$\begin{aligned} a' &= (1 + r)a + wz(1 - \ell) - c - T(wz(1 - \ell)) + Tr, && (38) \\ a' &\geq \underline{a}, && (39) \\ c &\geq 0, \qquad \ell \in [0, 1], && (40) \end{aligned}$$

where $V(a', z') = \max\{V^W(a', z'), V^E(a', z', e')\}$ incorporates the occupational choice.

4.2 Entrepreneur's Problem

An entrepreneur with state (a, z, e) and firm capital K solves

$$V^E(a, z, e) = \max_{c,\ell,K} \{u(c, \ell, G) + \beta_d \, \mathbb{E}[V(a', z', e')]\} \tag{41}$$

subject to the entrepreneur budget constraint (27) and the entry/exit conditions of Section 2.4.

4.3 Market Clearing

Definition 3 (Recursive Competitive Equilibrium)**.** *A recursive competitive equilibrium (RCE) consists of:*

1. *Value functions $V^W(a, z)$ and $V^E(a, z, e)$;*
2. *Policy functions $c(a, z)$, $\ell(a, z)$, and occupational choice $o(a, z, e)$;*
3. *Price functions $w(\Gamma)$ and $r(\Gamma)$;*
4. *A law of motion for the cross-sectional distribution $\mu_{t+1} = \Phi(\mu_t, \mathcal{C}_t)$;*
5. *A law of motion for the constitution $\mathcal{C}_{t+1} = \Psi(\mathcal{C}_t, \mu_t, w_t, r_t)$;*

such that:

(a) *Given prices and the constitution, agents optimize (solve their respective Bellman equations);*

(b) *Factor markets clear (Equations* (24) *and* (25)*);*

(c) *The goods market identity* (28) *holds;*

(d) *The distribution evolves consistently with individual decisions and shocks;*

(e) *The constitution evolves through the political process of Section 3.3, given the economic state.*

The key departure from standard RCE definitions is condition (e): the constitution is a state variable with an endogenous law of motion determined by agent proposals and votes.

5 Computational Method

5.1 Value Function Iteration

The worker's Bellman equation (37) is solved numerically via VFI on a discretized state space.

Asset grid. We construct an exponentially-spaced grid of $N_a = 100$ asset levels on $[\underline{a}, a_{max}]$ with $a_{max} = 1000$:

$$a_j = \underline{a} + (a_{max} - \underline{a}) \cdot \left(\frac{j}{N_a - 1}\right)^2, \qquad j = 0, 1, \ldots, N_a - 1. \tag{42}$$

The quadratic spacing concentrates grid points near the borrowing constraint $\underline{a}$, where the policy function has the highest curvature.

Leisure grid. A uniform grid of $N_\ell = 21$ leisure values: $\ell_k = k/20$ for $k = 0, 1, \ldots, 20$.

Consumption optimization. For each (a_j, z_s, ℓ_k) triplet, the budget constraint determines the maximum feasible consumption $c_{max} = budget - \underline{a}$ where $budget = (1 + r)a_j + wz_s(1 - \ell_k) - T(wz_s(1 - \ell_k)) + Tr$. We search over 11 equally-spaced consumption levels $c \in \{0, c_{max}/10, \ldots, c_{max}\}$.

Expected continuation value. The conditional expectation $\mathbb{E}[V(a', z') \mid z_s]$ is computed as

$$\mathbb{E}[V(a', z') \mid z_s] = \sum_{s'=1}^{N_z} \Pi(s, s') \cdot V^{interp}(a', s'), \tag{43}$$

where Π is the Rouwenhorst transition matrix and V^{interp} is the value function linearly interpolated on the asset grid.

Convergence. VFI iterates until $\|V_{n+1} - V_n\|_\infty < \epsilon_{VFI}$ with $\epsilon_{VFI} = 10^{-6}$, or a maximum of 500 iterations.

Remark 2 (Homogeneous Preference Speedup)**.** *When all agents share identical utility parameters (auto-detected at runtime), the VFI in Algorithm 1 is solved once, producing policy grids $c^*(a, z)$ and $\ell^*(a, z)$. Each agent's decision is then obtained by interpolation at their state (a_{it}, z_{it}), reducing the computational cost from $O(N)$ VFI solves to one.*

Remark 3 (NumPy Vectorization and Caching)**.** *The VFI implementation uses NumPy vectorized operations throughout, replacing nested Python loops with array computations for approximately 100× speedup. Budget arrays, utility evaluations, and expected value interpolation are computed as batch array operations. Additionally, VFI results are cached with keys based on rounded market parameters (wage, interest rate, public goods, transfer, tax rate*

Algorithm 1: Value Function Iteration

Input: Prices (w, r), public goods G, tax function $T(\cdot)$, transfer Tr, transition matrix Π

Output: Policy functions $c^*(a, z)$, $\ell^*(a, z)$

Initialize $V^0(a_j, z_s)$ for all (j, s);
for $n = 1, \ldots, 500$ **do**
    for $j = 1, \ldots, N_a;\ \ s = 1, \ldots, N_z$ **do**
        $V^{best} \leftarrow -\infty$;
        for $k = 0, \ldots, N_\ell$ **do**
            $\ell \leftarrow k/N_\ell$;
            $budget \leftarrow (1 + r)a_j + wz_s(1 - \ell) - T(wz_s(1 - \ell)) + Tr$;
            $c_{max} \leftarrow \max(0,\ budget - \underline{a})$;
            for $m = 0, \ldots, 10$ **do**
                $c \leftarrow c_{max} \cdot m/10$;
                $a' \leftarrow \max(budget - c,\ \underline{a})$;
                $EV \leftarrow \sum_{s'} \Pi(s, s') \cdot V^{n-1}_{interp}(a', s')$;
                $val \leftarrow u(c, \ell, G) + \beta_d \cdot EV$;
                if $val > V^{best}$ **then**
                    $V^{best} \leftarrow val$; $c^* \leftarrow c$; $\ell^* \leftarrow \ell$;
        $V^n(a_j, z_s) \leftarrow V^{best}$;
    if $\|V^n - V^{n-1}\|_\infty < 10^{-6}$ **then**
        break;
Interpolate (c^*, ℓ^*) at each agent's actual (a_{it}, z_{it});

proxy, and utility weights). When prices are nearly identical across consecutive periods, the cache avoids redundant VFI solves entirely.

5.2 Analytical Market Clearing

Equilibrium prices (w_t, r_t) are computed analytically from the Cobb-Douglas first-order conditions, yielding zero clearing error by construction.

Aggregate factor supplies are computed as:

$$L^s = \sum_{i:o_{it}=W} z_{it} \cdot \hat{h}_{it}, \qquad K^s = \max\left\{ \sum_{i=1}^{N} a_{it},\ \sum_{f \in \mathcal{F}_t} K_f \right\}, \tag{44}$$

where $\hat{h}_{it} = (1 - \ell_{it})$ if the household has a current labor supply decision, and $\hat{h}_{it} = 0.5$ as a fallback for stale labor supply values (i.e., before the first VFI solve). A floor guard of 10^{-8} is applied to both factor supplies to prevent division by zero.

With Cobb-Douglas production $Y = A_t(K^s)^\alpha(L^s)^{1-\alpha}$, the competitive equilibrium prices are:

$$w = (1-\alpha)\frac{Y}{L^s}, \qquad r = \alpha\frac{Y}{K^s} - \delta, \tag{45}$$

subject to the floor guards $w \geq 10^{-6}$ and $r \geq -0.99$.

Remark 4 (Zero Clearing Error)**.** *Because prices are derived directly from the aggregate production function and factor supplies, this analytical approach produces identically zero market clearing error ($\epsilon = 0$) each period. This eliminates the convergence uncertainty of iterative methods.*

Remark 5 (Historical Note)**.** *An earlier version of the model used a tatonnement (price adjustment) algorithm that iteratively updated prices to equate firm-level factor demands with aggregate supply. The analytical approach dominates: it is faster, simpler, and exact. Tatonnement remains available as a conceptual alternative for models with non-Cobb-Douglas production where closed-form solutions do not exist.*

5.3 Entrepreneurial Decision Solver

The entrepreneurial solver implements the utility-based occupational choice described in Section 2.4. The algorithm proceeds in four stages:

1. **Stationary distribution.** At initialization, compute the stationary distribution π^* of the ability Markov chain via power iteration (1000 iterations, tolerance 10^{-12}). The expected stationary ability is $\bar{e} = \sum_j \pi_j^* \cdot e_j$.

2. **Firm value.** For a given capital K, evaluate $V(e, K)$ using the closed-form formula (12): current-period profit at actual ability plus the discounted perpetuity of stationary expected profit.

3. **Optimal capital.** Find K^* via grid search over 20 equally-spaced points in $[\underline{K}, a_{it} - F]$, maximizing $V(e_i, K)$.

4. **Utility comparison.** Compute worker lifetime value V^W (13) and entrepreneur lifetime value V^E (14), including the entry cost penalty $\Phi(F)$ for new entrants. The agent enters entrepreneurship if $V^E > V^W$ and $a_{it} \geq \underline{K} + F$; an existing entrepreneur exits if $V^W > V^E$.

The solver also receives the current per-capita public goods level G as input, since G enters the utility function used for the occupational comparison.

Algorithm 2: Period Lifecycle (9-Step Algorithm)

Input: Previous period state $(t-1)$, configuration

Output: Updated period state t

Step 1: Draw shocks. Transition productivity and ability via Markov chains; draw aggregate TFP from AR(1); optionally draw preference shocks.;

Step 2: Clear markets. Find equilibrium (w_t, r_t) via analytical Cobb-Douglas FOCs (Section 5.2).;

Step 3: Collect decisions. Solve household Bellman via VFI (Algorithm 1); solve utility-based entrepreneurial entry/exit (Section 5.3), receiving per-capita public goods G for occupational comparison.;

Step 4: Validate constraints. Project decisions onto feasible set: clamp $\ell \in [0, 1]$, $c \leq budget - \underline{a}$, $a' \geq \underline{a}$.;

Step 5: Execute production. Compute firm outputs via (6); distribute income via (9); apply R&D shocks via (10); create/liquidate firms.;

Step 6: Enforce constitution. Apply tax rules, compute public goods, distribute transfers, apply regulations.;

Step 7: Process governance. Collect proposals, validate, vote, amend constitution (at proposal-interval periods). In benchmark mode, uses rule-based governance with dissatisfaction-driven proposals and value-weighted voting (Section 6.5).;

Step 8: Update states. Apply DSGE-HA budget constraint: $a' = (1+r)a + y + \pi - c - T + Tr$; compute realized utility $u(c, \ell, G)$.;

Step 9: Observe. Compute Gini, Pareto efficiency, aggregates (Y, C, I), wealth quantiles, welfare, firm stats; append to history.;

5.4 Period Lifecycle

Each period t executes nine steps in a fixed order, guaranteeing determinism.

Remark 6 (State Immutability). *Steps 1–9 operate on deep copies of the previous period's state. Agents never mutate global state directly; they return decisions that are validated and applied atomically in Step 8. This design prevents race conditions and ensures determinism (PROP-001).*

6 LLM-Augmented Political Decisions

6.1 Motivation

The model distinguishes between two types of decisions. *Economic decisions*—consumption, savings, labor supply—have well-defined objective functions (the Bellman equation) and can be solved numerically to arbitrary precision. *Political decisions*—what tax rate to propose, whether to add a means-tested transfer, how to vote on a minimum wage—are fundamentally open-ended. The space of possible proposals is large, the evaluation criteria are multi-dimensional (personal utility, ideological values, societal welfare), and the strategic interactions are complex.

This asymmetry motivates our design: analytical optimization for economics, LLM-augmented bounded rationality for politics. The LLM acts not as an optimizer but as a model of deliberative decision-making under genuine uncertainty about institutional design.

6.2 Architecture

The LLM decision engine operates through a five-stage pipeline:

1. **Context construction.** For each agent, build a structured JSON context containing: (a) personal state (wealth, productivity, role, utility parameters, value vector); (b) economic conditions (wage, interest rate, aggregate output); (c) current constitution (all rules and their parameters); (d) recent history (Gini trends, welfare changes, proposal outcomes); (e) a rule-type guide with examples and parameter descriptions.

2. **Cache lookup.** Compute a hash of each agent's context. If an identical context was seen earlier in the simulation, reuse the cached response. This avoids redundant API calls when multiple agents face similar conditions.

3. **Batched API calls.** Uncached contexts are batched (default batch size: 10) and sent to the LLM provider with structured output schemas. The LLM returns JSON conforming to the `PoliticalDecision` schema, which includes an optional proposal and a vote map.

4. **Schema validation.** Each LLM response is validated against a Pydantic model. Responses that fail validation trigger the fallback path.

5. **Fallback.** On API failure, timeout, or validation error, the agent defaults to a null political decision (no proposal, no votes), ensuring the simulation always progresses.

6.3 Information Structure

The political context provided to the LLM is richer than the economic context, reflecting the complexity of institutional design:

- **Societal conditions:** Gini coefficient, mean and median wealth, wealth quantiles (p10, p25, p50, p75, p90), unemployment rate, number of active firms, aggregate output, social welfare.

- **Recent history with trends:** The last 3–5 observation entries, each annotated with trend descriptors ("Inequality decreased," "Welfare improved," etc.) computed by comparing consecutive observations.

- **Recent proposal outcomes:** The last 20 vote outcomes, including whether proposals were passed, voted down, or rejected at validation, with vote counts. This enables the LLM to learn from prior political dynamics and avoid repeating failed approaches.

- **Rule-type guide:** A structured dictionary describing each of the eight rule types, their valid parameters, and example proposals. This serves as an instruction manual for the LLM's institutional design.

6.4 Determinism and Reproducibility

Several design choices ensure reproducibility:

- LLM temperature is set to 0, producing deterministic outputs for identical inputs.

- All randomness in agent selection, proposal ordering, and tie-breaking flows through a single seeded RNG.

- The cache maps contexts to responses deterministically.

6.5 Benchmark Mode

The model supports a *benchmark mode* that disables LLM calls entirely. In benchmark mode:

- Economic decisions are computed by the VFI solver.

- Entrepreneurial decisions are computed by the utility-based occupational choice solver.

- Governance uses rule-based decision-making via deterministic citizen logic.

Unlike earlier versions where the constitution remained static, benchmark mode now features a *rule-based governance* system. Each agent's political behavior is determined algorithmically:

- **Proposal gating.** Each agent skips proposal generation with probability 0.85, so that approximately 15% of agents propose in each governance period.
- **Dissatisfaction-based proposals.** Proposing agents evaluate their dissatisfaction with current constitutional rules based on their value vector (v^{eq}, v^{lib}). For instance, an agent with high v^{eq} is dissatisfied if taxes are low relative to their preferred rate $v^{eq} \times 0.5$. The agent proposes a modification to the rule with highest dissatisfaction, adjusting the parameter 30% toward their preferred value.
- **Value-weighted voting.** Each agent scores proposals using a weighted combination:

$$\text{Score} = 0.6 \times \text{value\_alignment} + 0.4 \times \text{self\_interest}, \tag{46}$$

  where value alignment measures how the proposal moves the rule toward the agent's ideological preference, and self-interest reflects the material impact (e.g., wealthier agents oppose higher taxes). The agent votes *for* if Score > 0.5.

This enables controlled experiments that compare LLM-augmented governance against a well-defined rule-based baseline, rather than a static constitution.

7 Calibration

Table 1 reports the baseline calibration. Parameter values are drawn from standard sources in the quantitative macro literature.

The capital share $\alpha = 0.33$ and depreciation rate $\delta = 0.10$ are standard in the macro literature. The productivity process parameters ($\rho_z = 0.9$, $\sigma_z = 0.2$) are in the range estimated by Storesletten et al. [9] for annual earnings. The entrepreneurial ability process is somewhat more volatile ($\sigma_e = 0.3$) and less persistent ($\rho_e = 0.85$), reflecting the higher turnover in business outcomes relative to labor earnings. The discount factor $\beta_d = 0.95$ and the Cobb-Douglas utility weights ($\alpha_u = 0.4$, $\beta_u = 0.35$, $\gamma_u = 0.25$) are within the range of estimated preferences in the literature on labor supply and public goods valuation. The entrepreneur leisure parameter $\ell_E = 0.2$ reflects the substantially higher time commitment of running a firm, and the firm discount factor $\beta_f = 0.95$ matches the household discount factor. The default tax rate of 10% provides a non-trivial initial fiscal environment, and the proposal skip probability of 0.85 ensures a moderate density of proposals in benchmark mode.

8 Discussion

8.1 Wealth Inequality Dynamics

The model generates endogenous wealth inequality through three channels. First, idiosyncratic productivity shocks create earnings inequality among workers. Second, entrepreneurial returns amplify the right tail of the wealth distribution, as successful entrepreneurs earn profits in addition to labor income. Third, the constitutional feedback loop creates a political economy of inequality: if wealthier agents successfully block redistributive proposals (exploiting their economic position to influence governance outcomes), inequality may be self-reinforcing.

The Gini coefficient is computed at each observation interval as

$$G = \frac{2\sum_{i=1}^{N} i \cdot a_{(i)} - (N+1)\sum_{i=1}^{N} a_{(i)}}{N\sum_{i=1}^{N} a_{(i)}}, \tag{47}$$

where $a_{(1)} \leq a_{(2)} \leq \cdots \leq a_{(N)}$ are ordered wealth levels.

8.2 Occupational Choice Mechanisms

The entry/exit dynamics are driven by the interaction of three factors:

1. **Ability:** High-ability agents have higher expected firm values, making entry more attractive.
2. **Wealth:** The minimum capital constraint $\underline{K}$ and entry cost F create a wealth barrier to entrepreneurship. Agents must satisfy $a_{it} \geq \underline{K} + F$ to enter.
3. **Prices:** Factor prices (w, r) affect both the opportunity cost of entrepreneurship (the forgone return $a \cdot r$ on savings) and the profitability of firms (through labor costs and capital costs).

The interaction of these factors generates rich dynamics. Rising wages reduce firm profitability, potentially triggering exit. A constitutional increase in tax rates reduces after-tax profits but may also reduce the opportunity cost of capital if it lowers interest rates through general equilibrium effects.

8.3 Constitutional Evolution

In LLM-augmented mode, the constitution evolves as agents observe economic outcomes and propose institutional changes. Several patterns emerge:

- **Tax ratchet:** If inequality is high, equality-preferring agents may propose higher taxes. If the tax rate becomes too high, liberty-preferring agents propose reductions. This creates oscillatory dynamics around an interior equilibrium.
- **Transfer design:** Agents may initially propose equal-share transfers, then shift to means-tested transfers as they observe that flat transfers have limited equalizing effect.
- **Voting rule changes:** Constitutional amendments to the voting threshold itself create meta-governance dynamics—agents may try to entrench favorable rules by raising the threshold required to change them.

8.4 LLM vs. Benchmark Comparison

Comparing LLM-mode and benchmark-mode simulations isolates the effect of LLM-augmented deliberation versus rule-based governance. In benchmark mode, the constitution evolves through rule-based governance with dissatisfaction-driven proposals and value-weighted voting (Section 6.5), starting from the default constitution (10% flat tax, equal-share transfers, majority voting). In LLM mode, constitutional changes are driven by LLM-generated proposals informed by rich economic context, potentially producing more nuanced institutional designs. Both modes feature endogenous governance, but they differ in the sophistication of the political reasoning.

The welfare comparison is measured by cumulative discounted social welfare:

$$W_T = \sum_{t=0}^{T} \bar{\beta}^t \sum_{i=1}^{N} u(c_{it}, \ell_{it}, G_t), \tag{48}$$

where $\bar{\beta}$ is the average discount factor across agents.

8.5 Pareto Efficiency

The Pareto efficiency score estimates the fraction of agent pairs for which no improving transfer exists. For each ordered pair (i, j) with $a_i \geq a_j$, we test whether a small transfer $\Delta = 1.0$ from i to j would improve j's utility without reducing i's:

$$u_j(c_j + \Delta) > u_j(c_j) \quad \text{and} \quad u_i(c_i - \Delta) \geq u_i(c_i). \tag{49}$$

The Pareto score is 1 minus the fraction of pairs where such an improvement exists.

9 Conclusion

We have developed a heterogeneous-agent model in which governance institutions are fully endogenous. Agents with heterogeneous productivity, ability, and preferences simultaneously make economic decisions (consumption, savings, labor, entrepreneurship) and political decisions (proposals, votes on constitutional rules). The economic and political layers are tightly coupled: constitutional rules determine fiscal policy, which enters budget constraints and affects equilibrium prices, which in turn shape the political preferences that generate new proposals.

The computational framework combines value function iteration for economic optimization with LLM-augmented decision-making for political deliberation. This asymmetric design reflects a genuine asymmetry in the underlying problems: economic optimization has well-defined structure (the Bellman equation), while institutional design is genuinely open-ended.

Several extensions merit future investigation. First, the model could incorporate a richer set of agent interactions, including coalition formation and bargaining. Second, the LLM's "preferences" could be calibrated against survey data on political attitudes. Third, the model could be scaled to larger populations using the Krusell-Smith algorithm for tracking the wealth distribution. Fourth, the constitutional framework could be extended to include judicial review and enforcement uncertainty.

A Rouwenhorst Method: Detailed Construction

We provide the full algorithm for the Rouwenhorst discretization of an AR(1) process $\log x_t = \rho \log x_{t-1} + \varepsilon_t$ with $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$.

Input: Persistence $\rho \in (0, 1)$, innovation volatility $\sigma > 0$, number of grid points $n \geq 2$.

Output: Grid values $\mathcal{S} = \{s_1, \ldots, s_n\}$ in levels, transition matrix $\Pi \in \mathbb{R}^{n \times n}$.

1. Unconditional standard deviation: $\sigma_x = \sigma / \sqrt{1 - \rho^2}$.

2. Grid bounds in logs: $z_{max} = \sigma_x \sqrt{n - 1}$.

3. Log grid: $z_j = -z_{max} + (j - 1) \cdot 2z_{max}/(n - 1)$ for $j = 1, \ldots, n$.

4. Transition parameter: $p = (1 + \rho)/2$.

5. Base matrix ($n = 2$):
$$\Pi_2 = \begin{pmatrix} p & 1-p \\ 1-p & p \end{pmatrix}.$$

6. Recursive construction. For $m = 3, \ldots, n$, define the $(m \times m)$ matrix $\widetilde{\Pi}_m$ by embedding Π_{m-1} in four quadrants (see (23)), then normalize rows to sum to 1.

7. Level grid: $s_j = \exp(z_j)$ for $j = 1, \ldots, n$.

The stationary distribution π^* satisfying $\pi^* = \pi^* \Pi$ is computed by initializing $\pi^{(0)} = (1/n, \ldots, 1/n)$ and iterating $\pi^{(k+1)} = \pi^{(k)} \Pi$ until convergence.

B VFI Convergence Properties

The VFI algorithm inherits convergence guarantees from the contraction mapping theorem. Define the Bellman operator $\mathcal{T}$:

$$(\mathcal{T}V)(a, z) = \max_{c,\ell} \left\{ u(c, \ell, G) + \beta_d \sum_{z'} \Pi(z, z') V(a', z') \right\}. \tag{50}$$

Under our assumptions (bounded utility, $\beta_d < 1$, compact choice set), $\mathcal{T}$ is a contraction mapping with modulus β_d in the sup-norm. By the Banach fixed-point theorem, $\mathcal{T}$ has a unique fixed point V^* and the sequence $V^n = \mathcal{T}^n V^0$ converges to V^* at rate β_d^n.

In practice, with $\beta_d = 0.95$, the error after n iterations is bounded by $\beta_d^n / (1 - \beta_d) \cdot \|V^1 - V^0\|_\infty$. For our tolerance of 10^{-6}, this typically requires 150–300 iterations depending on the state.

The discretization introduces approximation error from three sources:

1. **Asset grid:** The exponential spacing provides higher resolution near $\underline{a}$ where marginal utility is highest. Linear interpolation between grid points introduces $O(\Delta a^2)$ error.

2. **Leisure grid:** The uniform 21-point grid introduces at most $O(1/N_\ell)$ error in the leisure choice.

3. **Consumption grid:** The 11-point inner grid for consumption optimization provides a coarse approximation that could be refined at the cost of computation time.

C Full Parameter Table

Table 2 provides the complete parameter specification including derived and optional parameters.

D Sandbox Specification

Constitutional rules may include enforcement code—Python expressions that are evaluated in a sandboxed environment to compute taxes, transfers, or other policy outcomes. The sandbox imposes strict safety constraints:

Allowed AST nodes. Only the following Python AST node types are permitted:

- Literals (`Constant`)
- Arithmetic operators (`Add`, `Sub`, `Mult`, `Div`, `FloorDiv`, `Mod`, `Pow`)
- Unary operators (`USub`, `UAdd`)
- Comparison operators (`Eq`, `NotEq`, `Lt`, `LtE`, `Gt`, `GtE`)
- Boolean operators (`And`, `Or`, `Not`)
- Variable references (`Name`)
- Function calls (restricted to safe builtins)
- Conditional expressions (`IfExp`)
- Simple assignments (`Assign`)
- Container constructors (`Tuple`, `List`, `Dict`)

Disallowed constructs. Imports, function/class definitions, attribute access, comprehensions, lambda expressions, `exec`, `eval`, and all forms of I/O are blocked at the AST level.

Available builtins. The sandbox provides a restricted set of built-in functions: `abs`, `min`, `max`, `round`, `sum`, `len`, plus mathematical functions `sqrt`, `log`, `exp`, and `pow`.

Available variables. Enforcement code for tax rules receives `income`, `rate`, and `wealth`. Transfer rules receive `revenue` and `num_agents`. Public goods rules receive `fraction_of_revenue` and `revenue`.

This design ensures that constitutional rules, even when proposed by LLM agents, cannot compromise the simulation's integrity through arbitrary code execution.

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Table 1: Baseline Calibration

Parameter	**Symbol**	**Value**	**Description**
Core			
Number of agents	N	50	Population size
Time horizon	T	100	Number of periods
RNG seed	—	42	Reproducibility
Production			
Capital share	α	0.33	Cobb-Douglas
Depreciation rate	δ	0.10	Annual
Aggregate TFP (initial)	A_0	1.0	Normalized
Household preferences			
Consumption weight	α_u	0.40	Cobb-Douglas utility
Leisure weight	β_u	0.35	Cobb-Douglas utility
Public goods weight	γ_u	0.25	Cobb-Douglas utility
Discount factor	β_d	0.95	Intertemporal
Borrowing constraint	$\underline{a}$	0.0	Natural borrowing limit
Initial wealth mean	$\bar{a}_0$	100.0	Log-normal approx
Initial wealth std	σ_{a_0}	30.0	—
Idiosyncratic shocks			
Productivity persistence	ρ_z	0.90	AR(1)
Productivity volatility	σ_z	0.20	Innovation s.d.
Grid points	N_z	5	Rouwenhorst
Entrepreneurial			
Ability persistence	ρ_e	0.85	AR(1)
Ability volatility	σ_e	0.30	Innovation s.d.
Grid points	N_e	5	Rouwenhorst
Entry cost	F	5.0	Sunk cost
Minimum capital	$\underline{K}$	10.0	—
Firm discount factor	β_f	0.95	Perpetuity discounting
Entrepreneur leisure	ℓ_E	0.20	Reduced leisure (effort)
Aggregate shocks			
TFP persistence	ρ_A	0.95	AR(1)
TFP volatility	σ_A	0.01	Innovation s.d.
R&D			
Base success probability	p_0	0.10	—
TFP improvement mean	μ_{rd}	0.05	—
TFP improvement std	σ_{rd}	0.02	—
Market clearing			
Method	—	Analytical	Cobb-Douglas FOCs
Wage floor	—	10^{-6}	Numerical guard
Interest rate floor	—	-0.99	Numerical guard
VFI			
Asset grid points	N_a	100	Exponential spacing
Leisure grid points	N_ℓ	21	Uniform
Max iterations	—	500	—

Table 2: Complete Parameter Specification

Module	**Parameter**	**Default**	**Constraint**
Core	`num_agents`	50	≥ 20
Core	`max_periods`	100	≥ 1
Core	`seed`	42	integer
Shocks	`rho_z`	0.9	$(0, 1)$
Shocks	`sigma_z`	0.2	> 0
Shocks	`num_z_states`	5	$[2, 50]$
Shocks	`rho_a`	0.95	$(0, 1)$
Shocks	`sigma_a`	0.01	> 0
Shocks	`rho_e`	0.85	$(0, 1)$
Shocks	`sigma_e`	0.3	> 0
Shocks	`num_e_states`	5	$[2, 50]$
Shocks	`enable_preference_shocks`	False	boolean
Production	`alpha`	0.33	$(0, 1)$
Production	`delta`	0.1	$(0, 1)$
Production	`min_firm_capital`	10.0	> 0
Entrepreneurial	`firm_entry_cost`	5.0	> 0
Entrepreneurial	`firm_discount_factor`	0.95	$(0, 1)$
Entrepreneurial	`entrepreneur_leisure`	0.2	$(0, 1)$
Household	`a_min`	0.0	≥ 0
Household	`initial_wealth_mean`	100.0	> 0
Household	`initial_wealth_std`	30.0	≥ 0
Household	`homogeneous_preferences`	True	boolean
Household	`utility_alpha`	0.4	$(0, 1)$
Household	`utility_beta`	0.35	$(0, 1)$
Household	`utility_gamma`	0.25	$(0, 1)$
Household	`utility_beta_discount`	0.95	$(0, 1)$
Market clearing	`method`	analytical	Cobb-Douglas FOCs
Market clearing	`wage_floor`	10^{-6}	> 0
Market clearing	`interest_rate_floor`	-0.99	> -1
Intervals	`proposal_interval`	5	≥ 1
Intervals	`observer_interval`	5	≥ 1
Governance	`default_tax_rate`	0.1	$[0, 1]$
Governance	`proposal_skip_prob`	0.85	$[0, 1]$
LLM	`use_llm`	True	boolean
LLM	`llm_temperature`	0.0	$[0, 2]$
LLM	`llm_batch_size`	10	≥ 1
LLM	`llm_cache_enabled`	True	boolean
LLM	`benchmark_mode`	False	boolean
R&D	`rd_success_base_prob`	0.1	$[0, 1]$
R&D	`rd_tfp_improvement_mean`	0.05	> 0
R&D	`rd_tfp_improvement_std`	0.02	≥ 0